ISLAMIC UNIVERSITY OF TECHNOLOGY

DHAKA, BANGLADESH

ORGANIZATION OF ISLAMIC COOPERATION

DATABASE MANAGEMENT SYSTEMS LAB

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Report preparing for Lab Task

Task 1:

Create a user with username = <C_student_id> and password = cse4308 and grant necessary privileges to log in and execute DDL and DML statements. Then log in as that user.

Solution Approach:

```
CREATE USER C_210042132 IDENTIFIED BY cse4308 ;
GRANT ALL PRIVILEGES TO C_210042132;
CONNECT C_210042132 / cse4308 ;
```

SQL Command:

```
SQL> connect
Enter user-name: system
Enter password:
Connected.
SQL> CONNECT C_210042132 / cse4308 ;
Connected.
```

Task 2:

Write SQL statement to create a table 'INSTRUCTOR' which has 4 attributes:

- ID (assign it as Primary Key)
- NAME
- DEPT_NAME
- SALARY (ensure that SALARY is greater than 20000)

Solution Approach:

```
CREATE TABLE INSTRUCTOR
```

```
(
ID VARCHAR2(5) ,
NAME VARCHAR2 (20) NOT NULL ,
DEPT_NAME VARCHAR2(20) ,
SALARY NUMBER(5) ,
CONSTRAINT PK_IDENTITY PRIMARY KEY ( ID ),
CONSTRAINT SALARY_CHECK CHECK (SALARY > 20000 )
);
```

SQL Command:

```
SQL> CREATE TABLE INSTRUCTOR
  2  (
  3  ID VARCHAR2(5) ,
  4  NAME VARCHAR2 (20) NOT NULL ,
  5  DEPT_NAME VARCHAR2(20) ,
  6  SALARY NUMBER(5) ,
  7  CONSTRAINT PK_IDENTITY PRIMARY KEY ( ID ),
  8  CONSTRAINT SALARY_CHECK CHECK (SALARY > 20000 )
  9  );

Table created.
```

Task 3:

Write SQL statements to insert the following records into 'INSTRUCTOR' table:

ID	NAME	DEPT_NAME	SALARY
10101	Srinivasan	Comp. Sci.	65000
12121	Wu	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
00456	Gold	Physics	87000
45565	Katz	Comp. Sci.	75000

58583	Califieri	History	62000
76543	Singh	Finance	80000
76766	Crick	Biology	72000
03821	Brandt	Comp. Sci.	92000
98345	Kim	Elec. Eng.	80000

Solution Approach:

```
INSERT INTO INSTRUCTOR VALUES (’10101’ , ’Srinivasan’
, ’Comp.Sci.’ , 65000 );
INSERT INTO INSTRUCTOR VALUES (’12121’ , ’Wu’ ,
’Finance’ , 90000 );
INSERT INTO INSTRUCTOR VALUES (’15151’ , ’Mozart’ ,
’Music’ , 40000 );
INSERT INTO INSTRUCTOR VALUES (’22222’ , ’Einstein’ ,
’Physics’ , 95000 );
INSERT INTO INSTRUCTOR VALUES (’32343’ , ’El Said’ ,
’History’ , 60000 );
INSERT INTO INSTRUCTOR VALUES (’00456’ , ’Gold’ ,
’Physics’ , 87000 );
INSERT INTO INSTRUCTOR VALUES (’45565’ , ’Katz’ ,
’Comp. Sci.’ , 75000 );
INSERT INTO INSTRUCTOR VALUES (’58583’ , ’Califieri’
, ’History’ , 62000 );
INSERT INTO INSTRUCTOR VALUES (’76543’ , ’Singh’ ,
’Finance’ , 80000 );
INSERT INTO INSTRUCTOR VALUES (’76766’ , ’Crick’ ,
’Biology’ , 72000 );
INSERT INTO INSTRUCTOR VALUES (’03821’ , ’Brandt’ ,
’Comp. Sci.’ , 92000 );
INSERT INTO INSTRUCTOR VALUES (’98345’ , ’Kim’ ,
’Elec. Eng.’ , 80000 );
```

SQL Command:

```
Table created.

SQL> INSERT INTO INSTRUCTOR VALUES ('10101' , 'Srinivasan' , 'Comp.Sci.' , 65000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('12121' , 'Wu' , 'Finance' , 90000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('15151' , 'Mozart' , 'Music' , 40000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('22222' , 'Einstein' , 'Physics' , 95000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('32343' , 'El Said' , 'History' , 60000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('00456' , 'Gold' , 'Physics' , 87000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('45565' , 'Katz' , 'Comp. Sci.' , 75000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('58583' , 'Califieri' , 'History' , 62000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('76543' , 'Singh' , 'Finance' , 80000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('76766' , 'Crick' , 'Biology' , 72000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('03821' , 'Brandt' , 'Comp. Sci.' , 92000 );

1 row created.

SQL> INSERT INTO INSTRUCTOR VALUES ('98345' , 'Kim' , 'Elec. Eng.' , 80000 );

1 row created.
```

Task 4:

Write SQL statements to perform the following queries:

(a) Display all records of 'INSTRUCTOR' table.

(b) Show instructor ID and name only.

(c) Find name and department of instructors who have salary more than 70000.

(d) Find name and department of instructors who have salary in between 80000 and 10000

(inclusive).

(e) Find ID and name of instructors of Comp. Sci. department.

(f) Find name and salary of instructors of Finance department.

(g) Find ID and name of instructors of Comp. Sci. department or instructors who are paid

more than 75000.

(h) Find the names of the department.

Solution Approach:

a)

```
SELECT * FROM INSTRUCTOR ;
```

SQL Command:

```
SQL> set linesize 600;
SQL> set linesize 600
SQL> SELECT * FROM INSTRUCTOR ;

ID    NAME                                               DEPT_NAME            SALARY
----- -------------------------------------------------- -------------------- ----------
10101 Srinivasan                                         Comp. Sci.            65000
12121 Wu                                                 Finance               90000
15151 Mozart                                             Music                 40000
22222 Einstein                                           Physics               95000
32343 El Said                                            History               60000
456   Gold                                               Physics               87000
45565 Katz                                               Comp. Sci.            75000
58583 Califieri                                          History               62000
76543 Singh                                              Finance               80000
76766 Crick                                              Biology               72000
3821  Brandt                                             Comp. Sci.            92000

ID    NAME                                               DEPT_NAME            SALARY
----- -------------------------------------------------- -------------------- ----------
98345 Kim                                                Elec. Eng.            80000

12 rows selected.
```

b)

```
SELECT ID , NAME FROM INSTRUCTOR ;
```

SQL Command:

```
SQL> SELECT ID , NAME FROM INSTRUCTOR ;

ID    NAME
----- --------------------------------------------------
10101 Srinivasan
12121 Wu
15151 Mozart
22222 Einstein
32343 El Said
456   Gold
45565 Katz
58583 Califieri
76543 Singh
76766 Crick
3821  Brandt
```

c)

```
SELECT NAME , DEPT_NAME FROM INSTRUCTOR  WHERE SALARY >70000 ;
```

SQL Command:

```
SQL> SELECT NAME , DEPT_NAME FROM INSTRUCTOR  WHERE SALARY >70000 ;

NAME                                               DEPT_NAME
-------------------------------------------------- --------------------
Wu                                                 Finance
Einstein                                           Physics
Gold                                               Physics
Katz                                               Comp. Sci.
Singh                                              Finance
Crick                                              Biology
Brandt                                             Comp. Sci.
Kim                                                Elec. Eng.
```

d)

```
SELECT NAME , DEPT_NAME FROM INSTRUCTOR  WHERE 10000< SALARY AND SALARY  >80000 ;
```

SQL Command:

```
SQL> SELECT NAME , DEPT_NAME FROM INSTRUCTOR  WHERE 10000< SALARY AND SALARY  >80000 ;

NAME                                               DEPT_NAME
-------------------------------------------------- --------------------
Wu                                                 Finance
Einstein                                           Physics
Gold                                               Physics
Brandt                                             Comp. Sci.
```

e)

```
SELECT ID , NAME FROM INSTRUCTOR WHERE DEPT_NAME =  'Comp. Sci. ';
```

SQL Command:

```
SQL> SELECT ID , NAME FROM INSTRUCTOR WHERE DEPT_NAME =  'Comp. Sci. ';

ID    NAME
----- --------------------------------------------------
10101 Srinivasan
45565 Katz
3821  Brandt
```

f)

```
SELECT NAME ,SALARY FROM INSTRUCTOR  WHERE DEPT_NAME
= 'Finance '  ;
```

SQL Command:

```
SQL> SELECT NAME ,SALARY FROM INSTRUCTOR  WHERE DEPT_NAME = 'Finance '  ;

NAME                                                SALARY
-------------------------------------------------- ----------
Wu                                                   90000
Singh                                                80000
```

g)

```
SELECT ID , NAME FROM INSTRUCTOR WHERE DEPT_NAME
=  'Comp. Sci. ' OR SALARY >75000 ;
```

SQL Command:

```
SQL> SELECT ID , NAME FROM INSTRUCTOR WHERE DEPT_NAME =  'Comp. Sci. ' OR SALARY >75000 ;

ID    NAME
----- --------------------------------------------------
10101 Srinivasan
12121 Wu
22222 Einstein
456   Gold
45565 Katz
76543 Singh
3821  Brandt
98345 Kim
```

h)

```
SELECT DEPT_NAME  FROM INSTRUCTOR ;
```

SQL Command:

```
SQL> SELECT DEPT_NAME  FROM INSTRUCTOR ;

DEPT_NAME
--------------------
Comp. Sci.
Finance
Music
Physics
History
Physics
Comp. Sci.
History
Finance
Biology
Comp. Sci.

DEPT_NAME
--------------------
Elec. Eng.
```

Task 5:

Drop the 'INSTRUCTOR' table with all its constraints.

Solution Approach:

```
DROP TABLE INSTRUCTOR CASCADE CONSTRAINTS ;
```

SQL Command:

```
SQL> DROP TABLE STUDENT CASCADE CONSTRAINTS;

Table dropped.
```

Problems Faced:

1. Syntax Errors in CREATE TABLE Statements:

- Incorrect syntax in the CREATE TABLE statement resulted in table creation failures.

- Solution: Double-checking the syntax against SQL documentation or examples to ensure accuracy.

2. Data Integrity Issues in INSERT Statements:

- Providing incorrect values or omitting values in the INSERT statements resulted in data integrity issues.

- Solution: Thorough validation and accuracy checks when inserting data.

3. Handling NULL Values:

- Attributes with NULL values did not behave as expected in comparisons.

- Solution: Properly handling NULL values in queries and data processing.

4. Query Performance Optimization:

- Poorly optimized queries led to slow performance, especially with large datasets.

- Solution: Efficient use of indexes and optimization of complex queries.

5. Permanent Data Deletion:

- Dropping a table permanently removes its data.

- Solution: Exercise caution when performing such operations to prevent data loss.